

THE ROMANTICIZATION OF PROSTITUTION IN MOVIES

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1. INTRODUCTION

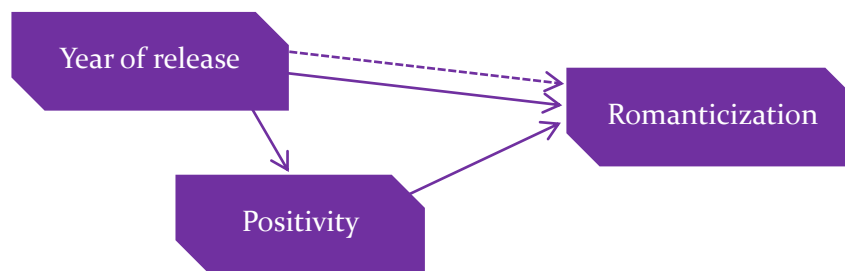
The recent release of the film *Anora* has called into question the status of prostitution in the film industry. Indeed, two pitfalls exist when addressing the subject of prostitution: the « oppression model » and the « celebratory model » (Wreiter, 2007). In the former, a phenomenalized and exaggerated victimization of prostitutes is emphasized, while in the latter, prostitution is romanticized.

In films in particular, the representation of prostitution can be inaccurate due to its glamorous or comic portrayal or due to incomplete representations of the world of prostitution (Blashdell, 2015).

In the post-war era, prostitution has been especially romanticized, particularly in literature (Harada, 2002).

Hence, this raises the question:

Did the romanticization of prostitution reach its peak in the post-war period, or does prostitution continue to be increasingly romanticized?



We anticipate that if this is the case, we should observe:

- **Prediction 1:** Movies depicting prostitution are generally more and more positive by the years.
 - **P2:** The more positive a movie (dealing with prostitution) is, the more romanticized its portrayal of prostitution is.
 - **P3:** Prostitution is more and more romanticized by the years.
- OR
- **P4:** The romanticization of prostitution reached its peak in the post war era, following a concave curve over time.

2. DATA

a. Data collection

The data comes from scrapping the IMDb (Internet Movie Database). They were collected by Pavan Kalyan Reddy Thota and are available on Kaggle. This first dataset contains around N=600 000 movies, with their name, genre, runtime, year, summary and other informations.

First, we reduced this dataset to around N = 300 000 movies by keeping only those with both a summary and a year of release available.

We then created a bag of words related to prostitution, and used it to select only movies dealing with this subject. This resulted in a dataset of N = 2708 movies dealing with prostitution topics, all of them having a year value and a summary text.

Eventually, we needed scale down our dataset because of the time needed for the LLM annotation. Hence, we sampled N = 800 movies from those 2708, trying to keep a uniform distribution of movies over time, in order to preserve enough variance in the years for our explanatory analysis.

b. Data annotation

We used the open source large language model (LLM) *phi-4* in order to annotate the data and create new variables.

For each movie, based on its title and summary, we asked the LLM to generate text and a numeric response associated to answer two questions. The first question dealt with the general estimated « positivity » or « negativity » of the movie. The scale went from -1, very negative movie, to 1, very positive movie. The second question dealt with the estimated romanticization of prostitution in the movie. The scale went from 0, no romanticization of prostitution, to 10, extreme romanticization of prostitution.

c. Data description

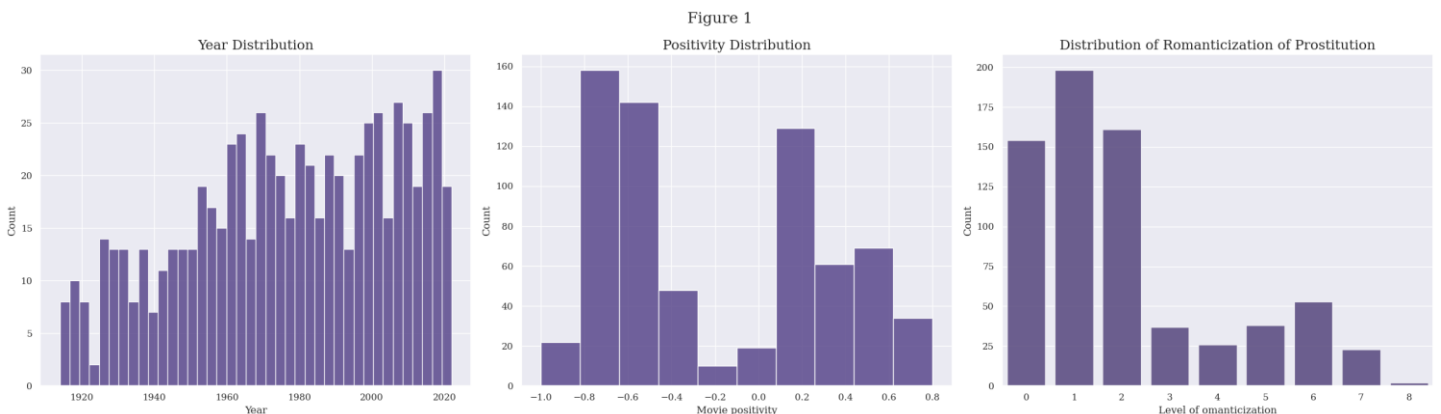


Fig. 1 : distribution our three values of interest, year of release of the movies, estimated movie positivity, estimated level of romanticization of prostitution.

3. METHODS

To test prediction 1, we performed a **simple linear regression** with the following formula (see **fig. 2** in the annex) :

$$\text{Movie positivity} \sim \text{Year of release of the movie}$$

To test prediction 2 and 3, we performed a **multivariate linear regression** with the following formula :

$$\text{Romanticization of prostitution} \sim \text{Positivity} + \text{Year}$$

To test prediction 2 and 4, we performed a **multivariate regression with quadratic effect** with the following formula :

$$\text{Romanticization of prostitution} \sim \text{Positivity} + \text{Year} + \text{Year}^2$$

4. RESULTS

The very slight effect of time over movie positivity ($\beta_1 = 0.0534$) is not significant ($p_1 = 0.139$). Moreover, the variance in the data is not at all explained by the model ($R^2 = 0.003$)

We cannot seem to explain the level of romanticization of prostitution based on the year of release of the movie. Indeed, time has almost no effect on the level of romanticization, whether in the linear ($\beta_2' = -0.0025$) or in the quadratic ($\beta_2'' = -0.0498$, $\beta_3'' = -0.0462$) model. Moreover, these effects are not significant ($p_2' = 0.338$, $p_2'' = 0.183$, $p_3'' = 0.176$).

However, romanticization of prostitution is significantly ($p_1' < 0.001$, $p_1'' < 0.001$) more likely to happen the more the movie is positive ($\beta_1' = 0.9393$, $\beta_1'' = 0.2292$) when controlled by time in our multivariate models.

Both the linear and quadratic models fail to fit the data and don't explain well the variance in our data ($R^{2'} = 0.054$, $R^{2''} = 0.056$). However, the quadratic still performs better compared to the linear ($AIC'' = 2110 < AIC' = 3223$). See fig. 4 in the annex for visualization of the coefficients.

MODEL	VARIABLE	COEFFICIENT	P-VALUE
<i>Positivity ~ Year</i> $R^2 = 0.003$	Year	$\beta_1 = 0.0534$	$p_1 = 0.139$

MODEL	VARIABLE	COEFFICIENT	P-VALUE
<i>Romanticization ~ Positivity + Year</i> $R^{2'} = 0.054$ $AIC' = 3223$	Positivity	$\beta_1' = 0.9393$	$p_1' < 0.001$
	Year	$\beta_2' = -0.0025$	$p_2' = 0.338$

MODEL	VARIABLE	COEFFICIENT	P-VALUE
<i>Romanticization ~ positivity + year + year²</i> $R^{2''} = 0.056$ $AIC'' = 2110$	Positivity	$\beta_1'' = 0.2292$	$p_1'' < 0.001$
	Year	$\beta_2'' = -0.0498$	$p_2'' = 0.117$
	Year ²	$\beta_3'' = -0.0462$	$p_3'' = 0.233$

Fig. 3 : table of results of the three models.

5. CONCLUSION

We have found a positive correlation between the general positivity of a movie and the romanticization of prostitution in it.

However, our results are not significant enough to reject the null hypothesis for the rest of our predictions. The level of romanticization in function of time doesn't seem to follow neither a linear nor a quadratic evolution. Nonetheless, the regression with quadratic effect seems to be slightly better model for our analysis.

6. REFERENCES

Weitzer, R. (2007). Prostitution: Facts and Fictions. *Contexts*, 6(4), 28-33. <https://doi.org/10.1525/ctx.2007.6.4.28>

Blasdel, R. (2015). « Reel or Reality? The Portrayal of Prostitution in Major Motion Pictures" *USF Tampa Graduate Theses and Dissertations*. <https://digitalcommons.usf.edu/etd/5912> (Pages 150 to 172)

Harada, K. (2002). *Good bad girls: male writers' romanticization of prostitutes in the post war era* (Doctoral dissertation, University of British Columbia). (Introduction)

7. ANNEX

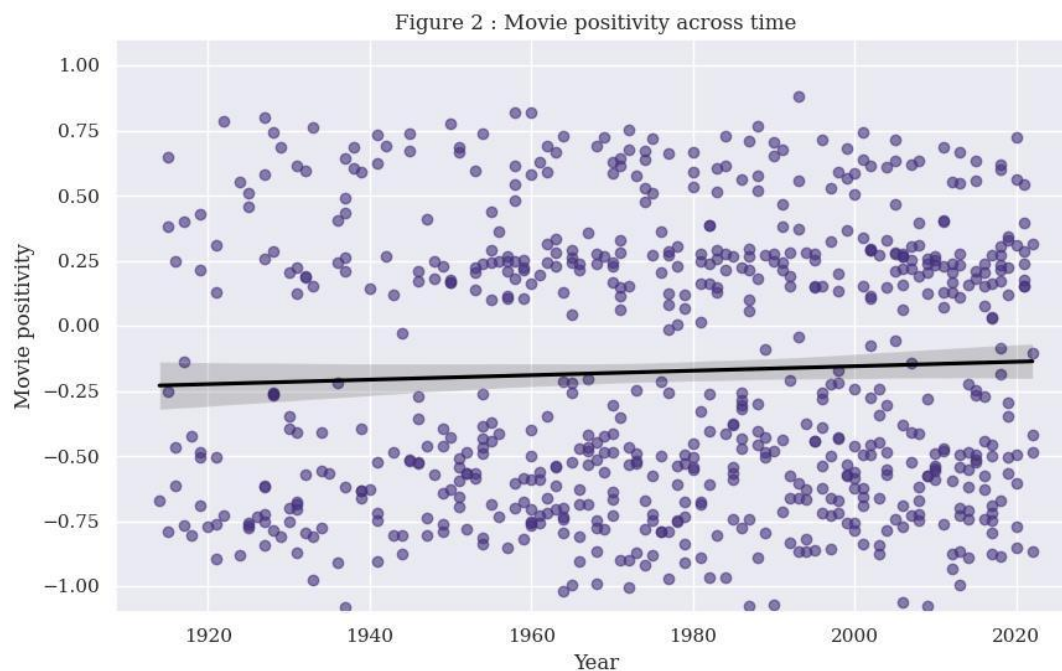


Fig. 2 : Scatter plot with regression line of estimated movie positivity in function of time (years). The plot uses the reel values but the model was fitted on standardized data.

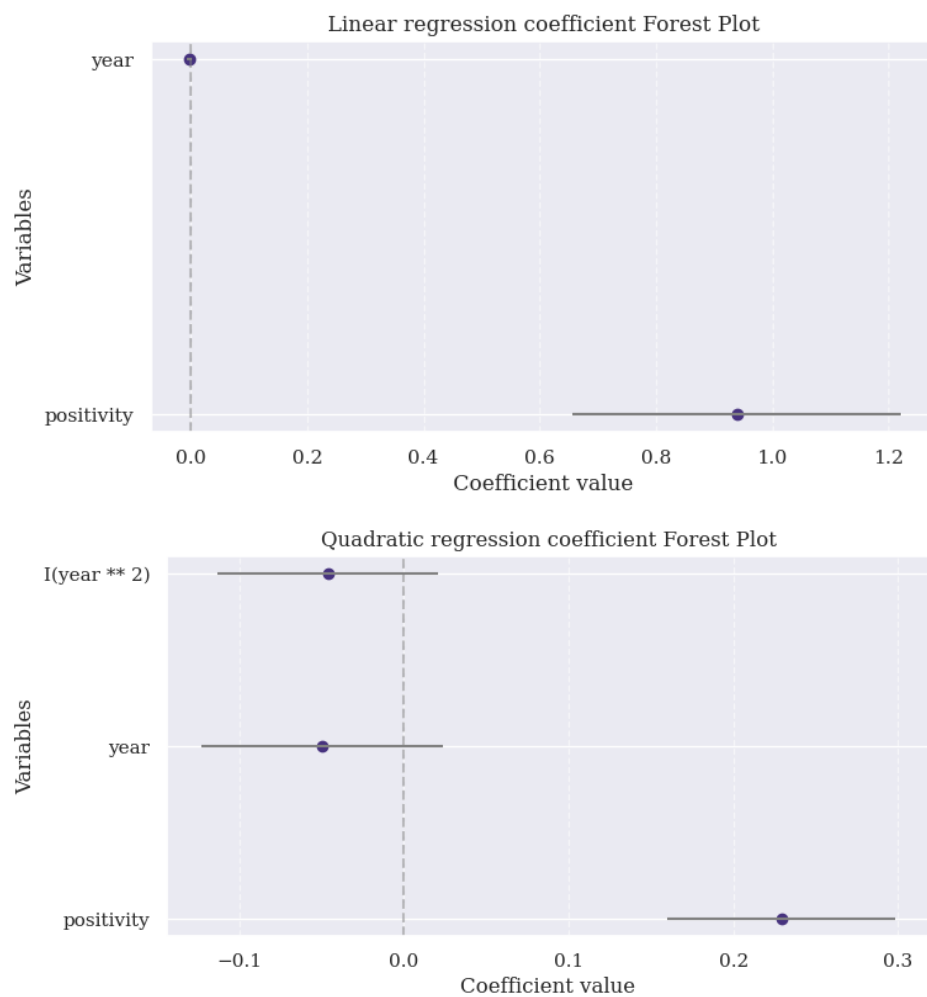


Fig. 4 : Forest plots of the coefficients in the multivariate regressions.